

Business Requirements Document (BRD)

Project Title: Library Management System (LMS)

Project Type: Full-stack Web Development Capstone Project

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Author: Kelvin Wong

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1. Document Purpose

This Business Requirements Document defines the functional and non-functional requirements for the development of a web-based Library Management System (LMS) that automates core library operations for librarians and provides a self-service portal for library members.

2. Project Overview

The Library Management System is a web application that streamlines library operations to allow members to reserve books online, borrow the book and pay fines. The system will serve two primary user groups: Librarians (Administrators) and Members.

3. Business Objectives

- Enable librarians to easily manage books and members.
- Provide a user-friendly interface for members to search for books and track their borrowing history.
- Create a robust and scalable system that can handle a moderate number of books and members.

4. Scope

4.1 In-Scope

- User authentication and role-based access control
- Member management
- Book catalogue management
- Borrowing, returning, reservation, and renewal of books
- Borrowing history and current loans tracking
- Automatic due-date tracking and fine calculation
- Password reset functionality

5. Acceptance Criteria

- All core features implemented and functional
- Code follows provided coding standards
- Basic test coverage
- Proper error handling
- Completed documentation
- No critical bugs

6. User Roles

Librarian (Admin):

- Library staff responsible for managing members, books, and lending transactions
- Full access to all modules

Member:

- Registered library patron
- Search books, view own borrowing history, reserve books, view own profile & fines

7. Functional Requirements

7.1 Authentication & User Management

ID	Requirement
AUTH-01	User registration (Members can self-register)
AUTH-02	Secure login with email/username and password
AUTH-03	Role-based access control (Admin / Member)
AUTH-04	Password reset via email link
AUTH-05	Profile view and edit (personal details)

7.2 Member Management (Librarian/Admin only)

D	Requirement
MEM-01	Create new member record with name, address, phone, email, membership start date
MEM-02	View, edit, and delete member records
MEM-03	Search members by name and member ID

7.3 Book Management

ID	Requirement
BOOK-01	Add new book with ISBN, title, author(s), category, publisher, publication year, total copies, copies available
BOOK-02	Update and delete existing books
BOOK-04	Search and filter books by title, author, ISBN, category, publication year
BOOK-05	Track real-time status: Available / Borrowed / Reserved

7.4 Lending Management

ID	Requirement
LEND-01	Issue (borrow) a book to a member (checks membership validity, borrowing limit, overdue items, fine threshold)
LEND-02	Record return of a book and calculate overdue fines if any
LEND-03	Allow members to reserve an available or currently borrowed book
LEND-04	Members can renew a borrowed book (max 2 renewals, no existing reservation by others)
LEND-05	View borrowing history (past and current loans) for members and librarians

8. Business Rules

8.1 Membership Rules

- Maximum 3 books may be borrowed at any time per member
- Membership is valid for 1 year from registration date
- Borrowing is blocked if membership has expired

8.2 Lending Rules

- Standard loan period: 14 days
- Maximum 2 renewals per book (each renewal adds another 14 days)
- Renewal not allowed if another member has reserved the book
- Member cannot borrow new books if they have any overdue items
- Member cannot borrow new books if total outstanding fines > \$10.00

8.3 Fine Policy

- Overdue fine: \$0.50 per day per book
- Fine calculation starts from the day after the due date
- Maximum fine per book: \$20.00 (fine stops accumulating after \$20)

9. Technical Stack (Defined for the Project)

9.1 Backend

- Framework: Spring Boot 3.x (Java 17+)
- Architecture: RESTful API
- Database: MySQL 8.x
- ORM: Spring Data JPA / Hibernate
- Security: Spring Security + JWT
- Build tool: Maven

9.2 Frontend

- Framework: React.js 18+
- State Management: React Hooks
- Routing: React Router v6
- UI Library: Material-UI (MUI) or Bootstrap 5
- HTTP Client: Axios
- Build tool: Vite or Create React App